Recipe 5

Filtering SAM files

Essential SAM concepts

A single line in a SAM file --> one alignment of a query sequence

A single line in SAM may also store information on a second query sequence (the "read mate")

But the read mate will also have line that contains contains the alignment information (so there is redundancy there)

One query may have multiple alignments (lines in the SAM file)!

Filtering SAM files

Filtering: generating a subset that only contains alignments with certain properties.

In a substantial number of studies **filtering** the most critical step.

Filtering process

```
samtools view
```

see help:

```
-f INT only include reads with all of the FLAGs in INT present [0]
-F INT only include reads with none of the FLAGS in INT present [0]
```

IMHO the wording is confusing... 😈

(there are more filtering commands)

samtools view

A more explicit formulation

`-f` keeps alignments that match **ALL** the flags
`-F` removes alignments that match **ANY** of the flags

You may use flags separately (perhaps a good practice)

```
samtools view -f 16 -f 64
```

Or merge the numbers into one `16 + 64 = 80`

```
samtools view -f 80
```

What is this flag 80?

```
samtools flags 80
```

prints

```
0x50    80      REVERSE,READ1
```

When do you use -f and when do you use -F?



It is a long and treacherous journey...

Build your flags one step at a time

Inclusion needs all flags to be on, rejection takes just one

Flags when combined with -f and -F may not end up mutually exclusive!

The opposite of something

IS "the something"!